

Sharp-exponent simplex stability with a quadratic dimension constant

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Abstract

For the lower-end projection-pyramid deficit, we prove centroid containment by every prescribed maximum-volume inscribed simplex with sharp exponent $1/(d-1)$ and an explicit coefficient at most $4096d^2$ in every dimension $d \geq 3$. The proof combines a conditioning-free, volume-weighted anchor theorem with an intrinsic first-moment conversion and a relative projection-cap argument. A truncated simplex gives the sharp obstruction, including the classification of all its maximum-volume inscribed simplices. The optimal coefficient lies between linear and quadratic order in the dimension; this paper does not close that gap. The traditional proof is complete. Existing Lean releases cover the original sharp-exponent upper theorem and the exact truncation obstruction; the improved quadratic coefficient is a written analytic refinement, not represented as fully formalized.

1 Definitions and the main statement

Let $K \subset \mathbb{R}^d$ be a full-dimensional compact convex body. Its projection body has support function $h_{\Pi K}(u) = \|u\|_2 \text{vol}_{d-1}(K|u^\perp)$. Write $R(K) = \text{vol}_d(\Pi K) / \text{vol}_d(K)^{d-1}$, and let $\mathcal{P}K$ be a pyramid in dimension $d+1$ with base K . Set

$$a(K) = \left(\frac{d}{d+1}\right)^d \frac{R(\mathcal{P}K)}{R(K)} - 1, \quad \mathfrak{e}(K) = a(K) - \frac{1}{d+1}.$$

The height and apex location do not affect this affine invariant. For an inscribed simplex S of maximum volume, let g_S denote its original centroid and define

$$E(K, S) = \inf\{t \geq 0 : K \subset g_S + (1+t)(S - g_S)\}.$$

Thus E is excess dilation, not the dilation factor itself. For a reference simplex Δ_d the affine Banach–Mazur distance to its affine class is

$$d_{\text{BM}}(K, \Delta_d) = \inf\{\lambda \geq 1 : S' \subset K \subset v + \lambda S', \text{ } S' \text{ a full-dimensional simplex, } v \in \mathbb{R}^d\}. \quad (1)$$

The free translation and optimization in (1) distinguish it from $E(K, S)$.

Theorem 1.1. *For every $d \geq 3$, every such K , and every prescribed maximum-volume inscribed simplex S ,*

$$\mathfrak{e}(K) \geq 0, \quad E(K, S) \leq G_d \mathfrak{e}(K)^{1/(d-1)},$$

$$G_d = 16(d+1)^2 [3d^2(d+1)^3(d+2)]^{1/(d-1)} \leq 4096d^2.$$

The exponent cannot be increased, even for the best choice of maximum S or for $d_{\text{BM}}(K, \Delta_d) - 1$. Any coefficient valid for every maximum S at this exponent is at least $(d+1)[(d+1)^2/(d(d-1))]^{1/(d-1)}$.

No product, join, smoothness or strict-convexity assumption is imposed on K . All maximum simplices exist by compactness. Replacement of one vertex of a maximum S by any $x \in K$ shows that every S -barycentric coordinate $\alpha_i(x)$ has $|\alpha_i(x)| \leq 1$. Consequently $E(K, S) \leq d + 1$.

2 Cone law and the exact deficit identity

Translate an interior point of K to zero, write $v = \text{vol}_d K$, and denote its surface-area measure by S_K . Push the probability measure $h_K(u) dS_K(u)/(dv)$ on the unit sphere forward by $u \mapsto u/h_K(u)$; call the resulting law ν . It is centered, bounded and supported on ∂K° , with full affine span. Centering follows from $\int u dS_K(u) = 0$; the total mass follows from the divergence formula. The support of S_K spans $\mathbb{R}^d$ for a full-dimensional body. Since ν is centered, linear and affine full span are equivalent. For independent samples put

$$A = \mathbb{E}|\det(X_1, \dots, X_d)|, \quad B = \mathbb{E}|\det((X_0, 1), \dots, (X_d, 1))|, \quad D = B - A.$$

Then $B > 0$ and

$$a(K) = \frac{B}{(d+1)A}, \quad \frac{D}{B} = \frac{(d+1)\epsilon(K)}{1 + (d+1)\epsilon(K)}.$$

Here is a direct derivation, including the normalization. For a polytope let u_i be facet area times outward unit normal and b_i facet area times support number. Set $H = \sum_{|I|=d} |\det(u_i : i \in I)|$ and $L = \sum_{|I|=d+1} |\det((u_i, b_i) : i \in I)|$. Cauchy's formula and the zonotope volume formula give $\text{vol} \Pi K = H$. For the pyramid with height t and apex above zero, the facet area vectors are $(0, -v)$ and $(tu_i/d, b_i/d)$, and its volume is $tv/(d+1)$. Therefore

$$\text{vol} \Pi(\mathcal{P}K) = (t/d)^d (vH + L/d), \quad a(K) = \frac{L}{dvH}. \quad (2)$$

Translation replaces b_i by $b_i - u_i \cdot z$, an elementary row operation, so (2) is translation invariant even when some uncentered b_i vanish. For the centered cone law, determinant multilinearity gives $A = d!H/(dv)^d$ and $B = (d+1)!L/(dv)^{d+1}$, proving the identity. General bodies follow by polytope approximation: choose a common interior ball and common exterior ball; support functions converge uniformly and surface-area measures weakly. The cone laws then converge weakly with uniformly bounded support, so both determinant integrals converge. Projection bodies and volumes also converge. Positivity of A, B follows from full span and positive measure neighborhoods of independent support tuples.

3 Conditioning-free volume-weighted anchors

The following statement is independent of convex-body realizability.

Lemma 3.1. *For any centered bounded Borel law with full affine support in $\mathbb{R}^d$, and any norm with support diameter Λ , there are affinely independent support anchors $w_0, \dots, w_d$ and a Borel assignment I such that*

$$\mathbb{E} \|X - w_{I(X)}\| \leq \Lambda \frac{(d+1)(d+2)}{2} \frac{D}{B}.$$

In particular $D \geq 0$.

Proof. For a base $\mathbf{x} = (x_1, \dots, x_d)$ put $F_{\mathbf{x}}(y) = \det((x_1, 1), \dots, (x_d, 1), (y, 1))$, $P(\mathbf{x}) = \mathbb{E}(F_{\mathbf{x}}(X))_+$ and $N(\mathbf{x}) = \mathbb{E}(-F_{\mathbf{x}}(X))_+$. Affinity and centering give $D = 2\mathbb{E} \min(P, N) \geq 0$, including singular bases. Define

$$\psi_{\mathbf{x}}(y, z) = \min((F(y))_+, (-F(z))_+) + \min((-F(y))_+, (F(z))_+).$$

For independent base and test points, each minimum has conditional mean at most $\min(P, N)$, so $\mathbb{E}\psi \leq D$. For an anchor tuple W write $V(W) = |\det((w_i, 1)_{i=0}^d)|$. Form $G(W, x)$ by summing the witness for each base of all anchors except w_i , tested at w_i, x , and for each $i < j$ the base of x and all anchors except w_i, w_j , tested at w_i, w_j . Orientations are irrelevant. There are $N_d = (d+1)(d+2)/2$ witnesses, each on $d+2$ independent samples. Thus for $H(W) = \int G(W, x) d\nu(x)$ we have $\mathbb{E}H \leq N_d D$.

When $V > 0$ let $\alpha_i(x)$ be anchor barycentric coordinates. The singleton witness is $V \min(1, (-\alpha_i)_+)$. The pair witness is $V[\min((\alpha_i)_+, (\alpha_j)_+) + \min((- \alpha_i)_+, (- \alpha_j)_+)]$. Choose the least index r maximizing α_i . Put $N_x = \sum_i (-\alpha_i)_+$ and $U_x = \sum_{i \neq r} (\alpha_i)_+$. Affinity and the independent diameter bound give

$$\begin{aligned} \|x - w_r\| &\leq \Lambda \min(1, N_x + U_x) \\ &\leq \Lambda \left[\sum_i \min(1, (-\alpha_i)_+) + \sum_{i < j} \min((\alpha_i)_+, (\alpha_j)_+) \right] \\ &\leq \Lambda G(W, x) / V(W). \end{aligned}$$

The middle inequality uses the pairs containing the largest positive coordinate to pay for U_x ; the capped negative terms pay for $\min(1, N_x)$. This remains valid for arbitrarily large negative coordinates.

Full affine support gives $B = \mathbb{E}V > 0$. Under the probability measure $d\hat{\mu} = (V/B)d\nu^{d+1}$, concentrated on $V > 0$,

$$\mathbb{E}_{\hat{\mu}}(H/V) = \mathbb{E}[H \mathbf{1}_{V>0}] / B \leq N_d D / B.$$

There exists a tuple in this probability-one domain with H/V at most its mean; otherwise the positive difference has positive integral. Selecting it and integrating the pointwise bound proves the lemma. This is not a minimum argument for a possibly singular ratio. The least-index assignment is Borel. If $D = 0$, its cost is zero and the law is supported on its $d+1$ anchors; their positive probabilities and centering put zero inside their simplex. $\square$

4 Intrinsic conversion to an enclosing simplex

We now translate the centroid of the prescribed maximum S to zero, and use this centering for the cone law. Write $n = d+1$, $Q = K - K$, $N(y) = h_Q(y)$, whose dual norm on u is the gauge $p_Q(u)$. Replacement maximality gives $-K \subset dS \subset dK$: the S -barycentric coordinates of $-x/d$ are $(1 - \alpha_i(x))/d$. For a cone support point, $h_K(x) = 1$, hence $N(x) \leq n$, and the support diameter is at most $2n$. Lemma 3.1 gives anchors and assignment of cost

$$h = \mathbb{E}N(X - w_I) \leq \frac{n^3(n+1)\epsilon}{1+n\epsilon} \leq C\epsilon, \quad C = (d+1)^3(d+2).$$

For nonzero u , every chord of K parallel to u has length at most the radial function of Q in that direction. Fiber integration and Cauchy's formula imply, with the homogeneous projection function π_K ,

$$\pi_K(u)/v \geq p_Q(u), \quad \mathbb{E}(u \cdot X)_+ = \pi_K(u)/(dv) \geq p_Q(u)/d.$$

Assignment error in that direction is at most $hp_Q(u)$, so $T = \text{conv}(w_i)$ contains $(1/d - h)Q^\circ$. When $h < 1/(2d)$, $0 \in \text{int } T$ and its polar simplex P satisfies $K \subset P \subset 2dQ$. Each w_i is on both polar boundaries and $h_K(w_i) = h_P(w_i) = 1$.

Let q_i be the vertex of P opposite the facet with normal w_i , and $\lambda_i > 0$ the barycentric coordinates of zero in T . The cone law of P has weights λ_i : they are its unique centered

probabilities. Its anchor barycentric functions are $\lambda_i(1 - q_i \cdot x)$. Set $p_i = \Pr(I = i)$ and $c = \sum p_i w_i$. Then $N(c) \leq h$ and

$$p_i - \lambda_i = -\lambda_i q_i \cdot c, \quad |q_i \cdot c| \leq 2dh =: r, \quad p_i \geq (1 - r)\lambda_i.$$

For $f_K(u) = \mathbb{E}|u \cdot X|$ and the analogous f_P we obtain

$$(1 - r)f_P(u) \leq \mathbb{E}|u \cdot w_I| \leq f_K(u) + hp_Q(u) \leq (1 + dh/2)f_K(u).$$

Thus $[\pi_P(u)/\text{vol } P]/[\pi_K(u)/v] \leq (1 + r/4)/(1 - r)$. Furthermore $z = \mathbb{E}[h_P(X) - 1] \in [0, r]$, because $h_P(y) \leq 2dN(y)$ and $h_P(w_I) = 1$. The mixed-volume formula gives $1 + z = V(K[d - 1], P)/v$, and Minkowski's first inequality yields $\text{vol } P/v \leq (1 + z)^d \leq (1 + r)^d$. Consequently

$$\delta := \sup_{u \neq 0} \frac{\pi_P(u) - \pi_K(u)}{\pi_P(u)} \leq 1 - \frac{1 - r}{(1 + r)^d(1 + r/4)} \leq (d + 5/4)r \leq 3d^2h.$$

The middle bound follows from $1 - abc \leq (1 - a) + (1 - b) + (1 - c)$ for factors in $[0, 1]$, and the derivative bounds for $(1 + r)^{-d}$ and $(1 + r/4)^{-1}$. Put $H = 3d^2C$. If $\epsilon \leq 1/H$, then $h \leq 1/(3d^2) < 1/(2d)$, $r < 1$ and $\delta \leq H\epsilon \leq 1$; every denominator and strict gate above remains valid at the endpoint.

5 Relative projection caps retain every maximum simplex

Lemma 5.1. *If $K \subset P$ and P is a simplex, put $\rho = [\sup_u (\pi_P(u) - \pi_K(u))/\pi_P(u)]^{1/(d-1)}$. Then every maximum $S \subset K$ satisfies $E(K, S) \leq 16n^2\rho$. If $\rho \leq 1/(16n)$, the stronger bound $E(K, S) \leq 8n\rho$ holds.*

Proof. For each vertex of P , a projection direction parallel to its opposite facet makes the corresponding barycentric coordinate descend to the projection. If $\max_K \lambda_i = 1 - t$, the missing projected cap has relative volume t^{d-1} . Hence choose $k_i \in K$ with $\lambda_i(k_i) \geq 1 - \rho$. Let Q_0 be their nonnegative column-stochastic barycentric matrix. Independent row choices from its columns show $\det Q_0 \geq 2 \prod_i (Q_0)_{ii} - 1 \geq 1 - 2n\rho$.

For $\rho \leq 1/(16n)$ put $\eta = 2n\rho \leq 1/8$. If W is the barycentric matrix of the prescribed maximum S , maximality gives $|\det W| \geq 1 - \eta$. Hadamard's inequality forces every column to have norm at least $1 - \eta$, hence an entry at least $(1 - \eta)^2 \geq 1 - 2\eta$. These entries have distinct rows: otherwise two columns collide with probability at least $(1 - 2\eta)^2$, and $|\det W| \leq \text{per } W \leq 4\eta < 1 - \eta$, a contradiction. Relabel so $W_{ii} \geq 1 - 4n\rho$. Then $\|W - I\|_1 \leq 8n\rho \leq 1/2$. For $B_0 = W^{-1}$,

$$\|B_0 - I\|_1 \leq \frac{8n\rho}{1 - 8n\rho}, \quad \beta := \max_{ij} |(B_0)_{ij} - \mathbf{1}_{i=j}| \leq \frac{4n\rho}{1 - 8n\rho} \leq 1/2.$$

The factor $1/2$ follows because columns of $B_0 - I$ have sum zero. Put $r_i = 1 - W_{ii}$. The ii entry of $B_0 W = I$ gives $(B_0)_{ii} - 1 \geq (1 - \beta)r_i/(1 - r_i) \geq r_i/2$. Replacing the i th vertex of S by k_i cannot increase volume, so $|(B_0 Q_0)_{ii}| \leq 1$. Its lower bound is $(1 - \rho)(B_0)_{ii} - \rho\beta$. Therefore $(B_0)_{ii} - 1 \leq \rho((B_0)_{ii} + \beta) \leq 2\rho$ and $r_i \leq 4\rho$. Now $\|W - I\|_1 \leq 8\rho$, so every negative entry of B_0 has magnitude at most $4\rho/(1 - 8\rho) \leq 8\rho$. These columns are precisely the S -barycentric coordinates of the vertices of P . A centroid homothety of excess t is characterized by all barycentric coordinates being at least $-t/n$. Thus $E(K, S) \leq 8n\rho$. For $\rho > 1/(16n)$ the universal bound $E \leq n$ gives $E < 16n^2\rho$. This proves both assertions without replacing the prescribed maximum S . $\square$

6 Completion of the upper bound and dimension dependence

On the branch $\epsilon \leq 1/H$, the enclosing simplex and Lemma 5.1 give $E \leq 16n^2(H\epsilon)^{1/(d-1)}$. On the other branch this same right side exceeds $n \geq E$. This proves the upper bound, including $\epsilon = 0$. For the explicit coefficient, $H_3 = 8640 < 144^2$, and

$$\frac{H_{d+1}}{H_d} = \left(\frac{d+1}{d}\right)^2 \left(\frac{d+2}{d+1}\right)^3 \frac{d+3}{d+2} \leq \frac{25}{6} < 144 \quad (d \geq 3).$$

Induction gives $H_d^{1/(d-1)} \leq 144$; since $n/d \leq 4/3$, $G_d \leq 4096d^2$. Also $G_d \sim 16d^2$. The local linear bound has an exponentially smaller deficit gate $H\epsilon \leq (16n)^{-(d-1)}$; it does not imply a global $O(d)$ coefficient.

A The truncation obstruction: complete calculation

Let $e_1, \dots, e_d$ be the coordinate vectors and, for $0 < t < 1$, set

$$K_t = \{x \in \mathbb{R}^d : x_i \geq 0, t \leq x_1 + \dots + x_d \leq 1\} = \text{conv}\{e_1, \dots, e_d, te_1, \dots, te_d\}. \quad (3)$$

This is a standard simplex with a homothetic cap cut from one vertex.

Theorem A.1. *For every integer $d \geq 2$ and $0 < t < 1$, the following assertions hold.*

(i) *With $q = t^{d-1}$, the invariant is exactly*

$$a(K_t) = \frac{1 + (d-1)q - (d-1)t^d - qt^d}{(1-t^d)(d+1+(d-1)q)}, \quad (4)$$

and consequently

$$\epsilon(K_t) = \frac{t^{d-1}[d(d-1) - (d+1)(d-2)t - 2t^d]}{(d+1)(1-t^d)(d+1+(d-1)t^{d-1})}. \quad (5)$$

In particular,

$$\lim_{t \downarrow 0} \frac{\epsilon(K_t)}{t^{d-1}} = c_d := \frac{d(d-1)}{(d+1)^2} > 0. \quad (6)$$

(ii) *The maximum simplex volume is*

$$M(K_t) = \frac{1-t}{d!}. \quad (7)$$

Every maximum-volume inscribed simplex, including those with vertices in the relative interiors of faces, has the form

$$S_p = \text{conv}\{e_1, \dots, e_d, tp\}, \quad p_i \geq 0, \quad \sum_{i=1}^d p_i = 1. \quad (8)$$

For each such simplex,

$$E(K_t, S_p) = (d+1)t \max_i p_i. \quad (9)$$

Thus

$$\inf_{S \text{ maximum}} E(K_t, S) = \frac{d+1}{d}t, \quad \sup_{S \text{ maximum}} E(K_t, S) = (d+1)t. \quad (10)$$

(iii) *The distance to any simplex satisfies*

$$\left(1 + t + \dots + t^{d-1}\right)^{1/d} \leq d_{\text{BM}}(K_t, \Delta_d) \leq 1 + t. \quad (11)$$

In fact, for every fixed maximum simplex S_p , its smallest enclosing translated homothet has factor exactly $1 + t$. In particular, for $0 < t < 1$,

$$\frac{t}{2d} \leq d_{\text{BM}}(K_t, \Delta_d) - 1 \leq t, \quad \liminf_{t \downarrow 0} \frac{d_{\text{BM}}(K_t, \Delta_d) - 1}{t} \geq \frac{1}{d}. \quad (12)$$

(iv) The Minkowski asymmetry is exactly

$$s(K_t) := \inf\{r \geq 1 : \exists z, -(K_t - z) \subset r(K_t - z)\} = d - t. \quad (13)$$

Proof. We prove each assertion directly.

Facet determinants. Put $c = 1/(d-1)!$ and $r = 1 - q$. The d coordinate facets have area $c(1 - t^{d-1})$. The top and bottom facets have areas $c\sqrt{d}$ and $c\sqrt{d}t^{d-1}$, respectively. The facet data in the coordinates (3) are therefore

$$(u_i, b_i) = (-cre_i, 0) \quad (1 \leq i \leq d), \quad (u_+, b_+) = (c\mathbf{1}, c), \quad (u_-, b_-) = (-cq\mathbf{1}, -ct^d). \quad (14)$$

The volume is $|K_t| = (1 - t^d)/d!$, so $d|K_t| = c(1 - t^d)$.

For horizontal d -minors, the all-coordinate subset contributes $c^d r^d$; the d subsets with the top facet contribute $c^d r^{d-1}$ each; and the d subsets with the bottom facet contribute $c^d q r^{d-1}$ each. Subsets with both top and bottom have determinant zero because their normals are parallel. Hence the horizontal sum is

$$H = c^d r^{d-1} [d + 1 + (d-1)q]. \quad (15)$$

For the lifted $(d+1)$ -minors, the subset containing all coordinate facets and the top facet contributes $c^{d+1} r^d$. Replacing the top facet by the bottom facet gives $c^{d+1} r^d t^d$. Each of the d subsets with top, bottom and $d-1$ coordinate facets contributes $c^{d+1} r^{d-1} (q - t^d)$; here $q - t^d = q(1 - t) > 0$. Thus the lifted sum is

$$L = c^{d+1} r^{d-1} [1 + (d-1)q - (d-1)t^d - qt^d]. \quad (16)$$

Dividing L by $d|K_t|H$ gives (4); subtracting $1/(d+1)$ gives (5). The bracket in (5) decreases strictly to zero as $t \uparrow 1$, so the defect is positive. Dividing (5) by t^{d-1} and passing to $t = 0$ proves the exact limit (6), without any numerical fitting.

Maximum simplices, including nonvertex ones. First consider a simplex selected from the $2d$ vertices in (3). If a ray $\mathbb{R}_+ e_i$ is unrepresented, the simplex lies in the coordinate hyperplane $x_i = 0$ and has zero volume. Otherwise $d+1$ distinct vertices on d rays give exactly one doubled ray, with vertices e_j, te_j , and one vertex on every other ray. Subtract the two doubled columns in the lifted vertex determinant. If k of the other $d-1$ selected vertices are bottom vertices, its absolute determinant is

$$(1 - t)t^k. \quad (17)$$

The largest value is $1 - t$, attained precisely by all d top vertices and one bottom vertex. Repeated vertices give determinant zero.

Now let $z_0, \dots, z_d$ be arbitrary vertices of an inscribed simplex. Represent each z_j as a convex combination of the $2d$ vertices, and independently sample a vertex Y_j from that combination. Multiaffinity of the lifted determinant D gives

$$|D(z_0, \dots, z_d)| = |\mathbb{E}D(Y_0, \dots, Y_d)| \leq \mathbb{E}|D(Y_0, \dots, Y_d)| \leq 1 - t. \quad (18)$$

This proves the upper bound (7), already attained by the preceding examples. If the simplex is maximum, equality in (18) implies that $|D(Y_0, \dots, Y_d)| = 1 - t$ almost surely. By (17),

exactly one of the Y_j is a bottom vertex and the other d are the d distinct top vertices, almost surely.

Let I_j be the indicator that Y_j is a bottom vertex. Independence and $\sum_j I_j = 1$ almost surely imply

$$0 = \text{Var}\left(\sum_j I_j\right) = \sum_j \mathbb{P}(I_j = 1)[1 - \mathbb{P}(I_j = 1)].$$

Thus every slot has deterministic top/bottom type, with precisely one bottom slot. The supports of the d top slots must be pairwise disjoint; otherwise independent sampling gives a repeated top vertex with positive probability. There are d nonempty disjoint supports in a set of d top vertices, so each is a singleton. The bottom slot is any convex combination tp . This proves (8). Conversely, the height above the hyperplane $\sum x_i = 1$ of any point tp is fixed, so every simplex (8) has volume $(1 - t)/d!$.

Centroid containment. Write $s_x = \sum_i x_i$. Barycentric coordinates for S_p , with index 0 for tp , are

$$\beta_0(x) = \frac{1 - s_x}{1 - t}, \quad \beta_i(x) = x_i - tp_i \beta_0(x) \quad (1 \leq i \leq d). \quad (19)$$

Evaluation at the $2d$ vertices of K_t shows

$$\min_{K_t} \beta_0 = 0, \quad \min_{K_t} \beta_i = -tp_i. \quad (20)$$

For the latter minimum, use te_j with $j \neq i$, available since $d \geq 2$. Affine functions attain their extrema on these vertices. The centroid-dilated simplex $g_{S_p} + (1 + \epsilon)(S_p - g_{S_p})$ is exactly the region $\beta_i \geq -\epsilon/(d + 1)$ for every $0 \leq i \leq d$. Equations (20) give (9), and optimizing $\max_i p_i$ on the probability simplex gives (10).

Translated containment and distance to any simplex. For barycentric coordinates of any simplex S , a translated homothet $v + \lambda S$ has inequalities $\beta_i \geq a_i$ with $\sum_i a_i = 1 - \lambda$. Conversely any such thresholds describe a translated homothet. To contain K , one needs $a_i \leq \min_K \beta_i$ for each i , so

$$\lambda \geq 1 - \sum_i \min_K \beta_i. \quad (21)$$

Choosing all thresholds equal to their minima realizes equality whenever $S \subset K$. For (20), the sum is $-t$, giving factor exactly $1 + t$. Concretely,

$$K_t \subset -tp + (1 + t)S_p. \quad (22)$$

This proves the upper bound in (11) under the full freedom in (1).

For the lower bound, any simplex $S \subset K_t$ has $|S| \leq M(K_t)$. If $K_t \subset v + \lambda S$, volume monotonicity and (7) imply

$$\lambda^d \geq \frac{|K_t|}{|S|} \geq \frac{|K_t|}{M(K_t)} = \frac{1 - t^d}{1 - t} = 1 + t + \dots + t^{d-1}. \quad (23)$$

Taking the infimum proves the lower bound in (11). Furthermore

$$(1 + t)^{1/d} - 1 = \frac{1}{d} \int_0^t (1 + u)^{-(d-1)/d} du \geq \frac{t}{2^{(d-1)/d}} \geq \frac{t}{2d}.$$

Finally the derivative at zero of $(1 + t + \dots + t^{d-1})^{1/d}$ is $1/d$, proving (12). This lower bound concerns *all* simplices, not only maximum simplices. We do not assert an exact formula for the unrestricted distance in (11).

Minkowski asymmetry. The containment in (13) is equivalent to $-K_t + (r+1)z \subset rK_t$. Writing $Z = \sum_i z_i$, its necessary and sufficient facet inequalities are

$$(r+1)z_i \geq 1 \quad (1 \leq i \leq d), \quad 1 + rt \leq (r+1)Z \leq r+t. \quad (24)$$

Summing the coordinate inequalities and using the last upper bound gives $r \geq d-t$. This value is attainable: take $z_i = 1/(d+1-t)$. The bottom inequality follows from

$$d-1 - (d-t)t = (1-t)(d-1-t) \geq 0.$$

All other inequalities in (24) are equalities. This proves (13). $\square$

B Universal implications and necessary constants

Corollary B.1. *Fix $d \geq 3$. A power estimate valid for every full-dimensional body of dimension d , even restricted to a sufficiently small lower-end defect, can have exponent α only if $\alpha \leq 1/(d-1)$, for each of:*

$$d_{\text{BM}}(K, \Delta_d) - 1 \leq C_d \mathfrak{e}(K)^\alpha; \quad (25)$$

$$\inf_{S \text{ maximum}} E(K, S) \leq C_d \mathfrak{e}(K)^\alpha; \quad (26)$$

$$E(K, S) \leq G_d \mathfrak{e}(K)^\alpha \quad \text{for every maximum-volume inscribed simplex } S. \quad (27)$$

At the critical exponent $1/(d-1)$, any constant in (27) must satisfy

$$G_d \geq (d+1) \left(\frac{(d+1)^2}{d(d-1)} \right)^{1/(d-1)} > d+1. \quad (28)$$

Any constant in (26) must be at least $1/d$ times the right-hand side of (28), and any constant in (25) must be at least

$$\frac{1}{d} \left(\frac{(d+1)^2}{d(d-1)} \right)^{1/(d-1)}. \quad (29)$$

These are necessary constants, not claimed optimal universal constants. In particular a dimension-independent constant is impossible in (27) at the critical exponent.

Proof. Equations (6), (10), and (12) give a positive constant times t on the left of each estimate and a constant times $t^{(d-1)\alpha}$ on its right. If $(d-1)\alpha > 1$, division by t and passage to $t \downarrow 0$ is a contradiction. At equality of exponents, take $p = e_1$ in (9) for (28), $p_i = 1/d$ for (26), and the lower-limit bound in (12) for (29). $\square$

C Provenance, formal scope and prior work

The invariant is taken from mxym/math entry 005, whose facet and cone identities are derived above. The volume-weighted anchor argument, intrinsic conversion and relative cap bridge consolidate the later research packages; the full source map and fixed hashes accompany this manuscript. The truncation appendix retains the original complete computation and all-maximum-simplex classification. Together they prove the remaining sharpness assertions in Theorem 1.1. Its formulas are valid also for $d = 2$, while the main upper theorem is stated for $d \geq 3$.

The frozen releases `formalizations/sharp-simplex-upper-bound` and `formalizations/simplex-truncation-sharpness` provide Lean proofs of the earlier sharp-exponent bound and the truncation obstruction, respectively. The former has a larger coefficient. They must

not be described as Lean proofs of the present $G_d \leq 4096d^2$ refinement. Fresh compilation, target axiom checks and empty-environment kernel replay have distinct verification records; arithmetic regressions do not prove this analytic refinement. Full formalization of the improved coefficient and optimal dimension dependence remain open tasks. No external human review or literature-wide priority certification is claimed.

Classical tools used above are Cauchy’s projection formula, zonotope volumes, the surface-area and mixed-volume formulas, Minkowski’s first inequality, polytope approximation and Hadamard’s inequality. Standard references are R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd edition, Cambridge University Press, 2014, and P. McMullen, *On zonotopes*, Transactions of the AMS 159 (1971), 91–109. These are inputs, not project results.

Related results include K. J. Böröczky Jr., *The stability of the Rogers–Shephard inequality and of some related inequalities*, Advances in Mathematics 190 (2005), 47–76, DOI 10.1016/j.aim.2003.11.015. In the plane, the invariant identifies with a difference-body ratio and the published result gives a direct linear Banach–Mazur comparison. It does not directly give the prescribed-centroid, all-maximizer assertion in dimensions $d \geq 3$. The accompanying limited comparison also records cone-volume and random-simplex stability papers. No absence of an exact match in that screen is treated as proof of novelty. The exact deficit, prescribed centroid and universal quantifier over maximum simplices are essential when comparing statements.